



bp India Student Challenge 2025-26

Reservoir Revivers

Where Insight Revives Energy

PRODUCTION HISTORY EVALUATION



P01→W01+W02

Laurie Segment SE
background channel
complex

P02→W03+W04

Laurie Segment SE
High Net Channel

P03→W05+W06

Laurie Segment NE
High Net Channel

P04→W06+W08

Laurie Segment NE
background channel
complex

P05→W07

Laurie Segment NE
background channel
complex



Laurie Reservoir Roadmap



Fig.1: Field-wide injection and production rate trends.

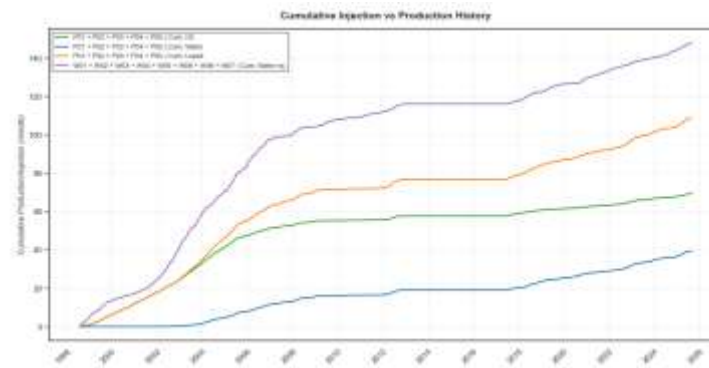


Fig.2: Cumulative oil, water, liquid production, and water injection.

Grouped injector-producer pairs by lithofacies because the specific rock architecture connecting them whether highly permeable 'highways' or baffled mud zones-directly controls pressure communication, sweep efficiency, and the timing of water break-through.

EXECUTIVE SUMMARY: MACRO-FIELD BEHAVIOUR OF PRODUCTION AND INJECTION WELLS

First injection date: July 1998

First Production date: August 1998

Peak oil rate: 29.99 mbd (Dec 2003)

Peak water rate: 23.75 mbd (May 2012)

Peak liquid rate: 43.01 mbd (May 2012)

Peak injection rate: 70.46 mbd (July 2002)

Cumulative oil produced: 69.25 MMstb

Cumulative water produced: 39.48 MMstb

Cumulative liquid produced: 108.73 MMstb

Cumulative water injected: 147.98 MMstb

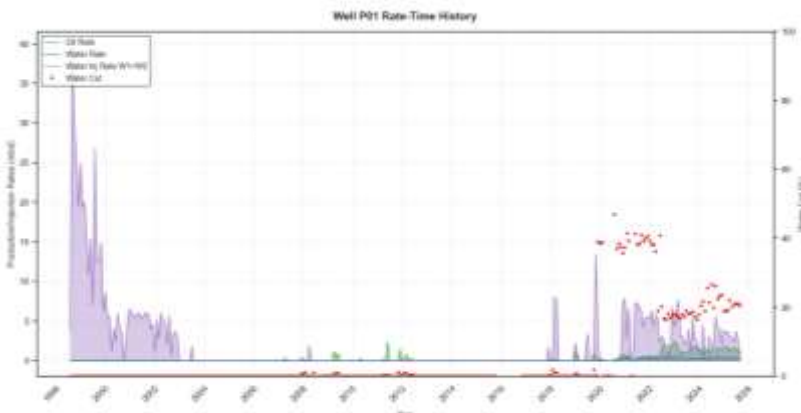
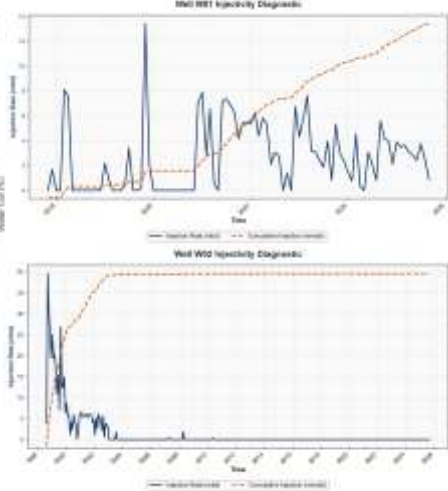


Fig.3: Production and water cut profile for Well P01, paired with individual injectivity diagnostics for supporting injector wells W01 and W02.



Parameter	INTERPRETATION OF P02
Well Performance	High initial oil → water breakthrough spike (fractures) → long-term high water cut with bypassed oil
Injectors Associated	W04 (main), later W03
Producer-Injector Relationship	Strong connectivity but over-efficient due to fractures → short-circuiting
Pressure Support Pattern	Initially good → later inefficient (water cycling through fractures)
Reservoir Productivity (Spatial)	Located in thickest part of High Net channel (axial) → very high potential
Lithofacies Control	PF1 amalgamated sands → excellent permeability but prone to channeling
Key Geological Controls	Compactional high + nearby muddy channel → compartmentalization + bypassed oil zones

Parameter	INTERPRETATION OF P01
Well Performance	Initially no production → later high water cut, low oil → finally high oil, low water cut (after lower zone injection)
Injectors Associated	W02 (early), W01 (upper → later lower zone)
Producer-Injector Relationship	Initially poor connectivity (fault leakage, basal scour) → later vertical zonal control critical
Pressure Support Pattern	Early: No pressure support (water lost to thief zones) → Later: Effective support when targeting lower sands
Reservoir Productivity (Spatial)	Located in background channel (off-axis) → moderate productivity, not as good as axial wells
Lithofacies Control	PF2/PF3 (heterolithic) → baffles → water fingering ; PF1 (amalgamated sands) → good sweep efficiency
Key Geological Controls	Fault + basal scour → vertical communication; heterogeneity → selective sweep

The **sinusoidal water cut** response in **Well P02** suggests strong **inter-well connectivity** through **high-conductivity streaks**, requiring reduced injection rates to mitigate early water breakthrough and maintain conformance control.

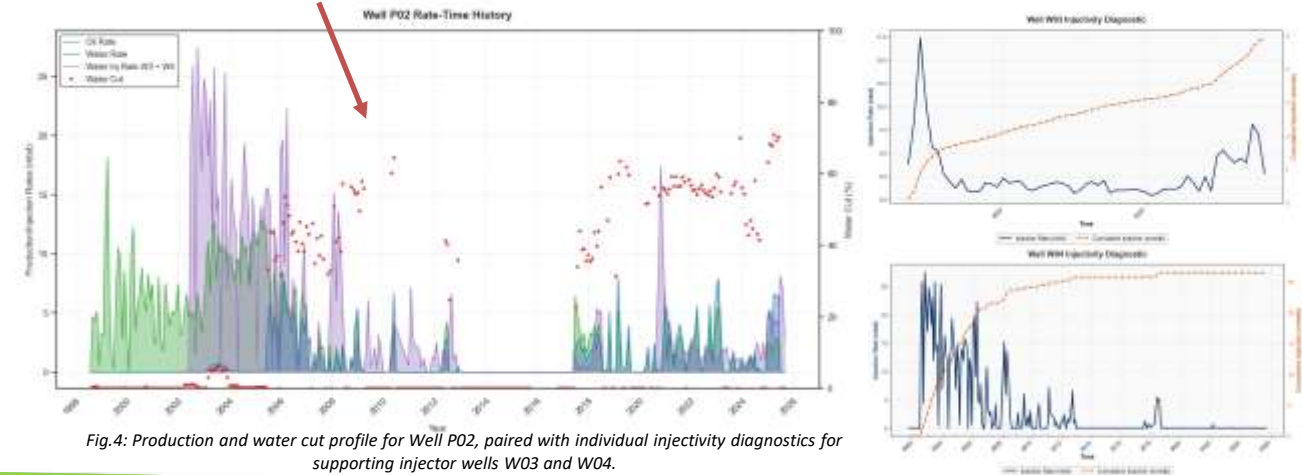


Fig.4: Production and water cut profile for Well P02, paired with individual injectivity diagnostics for supporting injector wells W03 and W04.

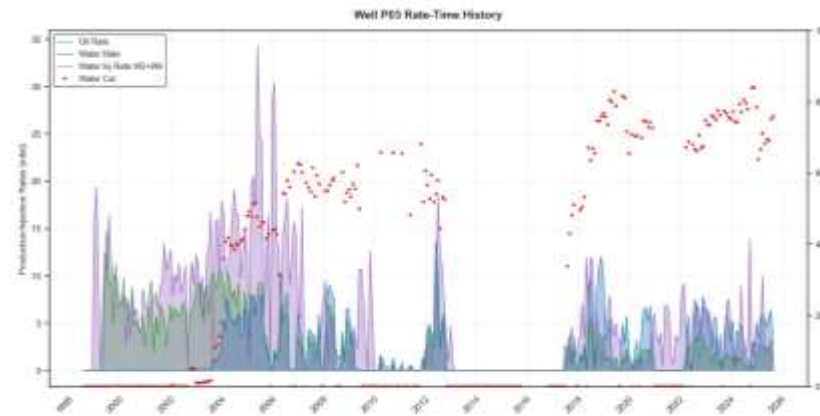
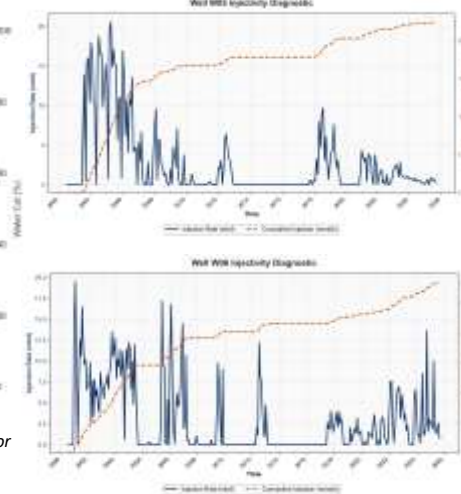


Fig.5: Production and water cut profile for Well P03, paired with individual injectivity diagnostics for supporting injector wells W05 and W06.

Parameter	Interpretation of P03
Well Performance	Extremely high oil initially → cyclic oil spikes → eventual very high water cut (>90%)
Injectors Associated	W05, W06
Producer-Injector Relationship	Excellent connectivity (axial sands) → eventually too efficient → direct water cycling
Pressure Support Pattern	Initially strong → later over-support → crest flooding
Reservoir Productivity (Spatial)	Located at structural crest (attic oil zone) → best productivity initially
Lithofacies Control	PF1 amalgamated sands → very high permeability ; P6 mud baffles → delay water coning
Key Geological Controls	Structural high + top seal + axial connectivity



Parameter	Interpretation of P04
Well Performance	Low oil → erratic → chaotic slugging → eventual abandonment
Injectors Associated	W06
Producer-Injector Relationship	Weak initially → later localized breakthrough through thief paths
Pressure Support Pattern	Initially excessive (high VRR) → later inefficient due to baffles + channeling
Reservoir Productivity (Spatial)	Located in heterolithic sands → low productivity
Lithofacies Control	Thin interbedded sand-shale → flow barriers + tortuous flow paths
Key Geological Controls	Baffles → pressure buildup → sudden breakthrough → thief zone creation

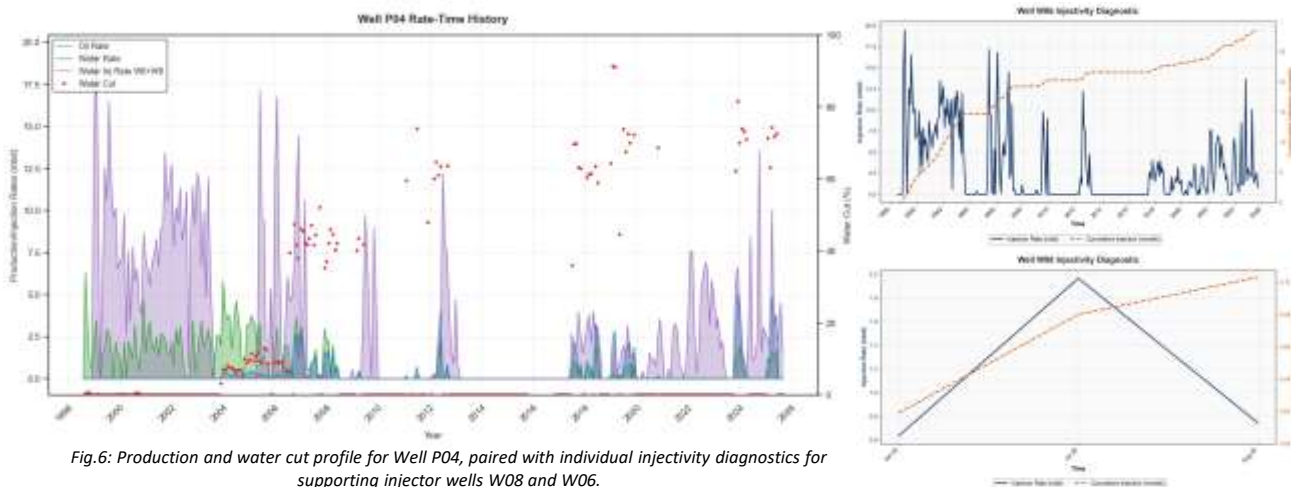


Fig.6: Production and water cut profile for Well P04, paired with individual injectivity diagnostics for supporting injector wells W08 and W06.

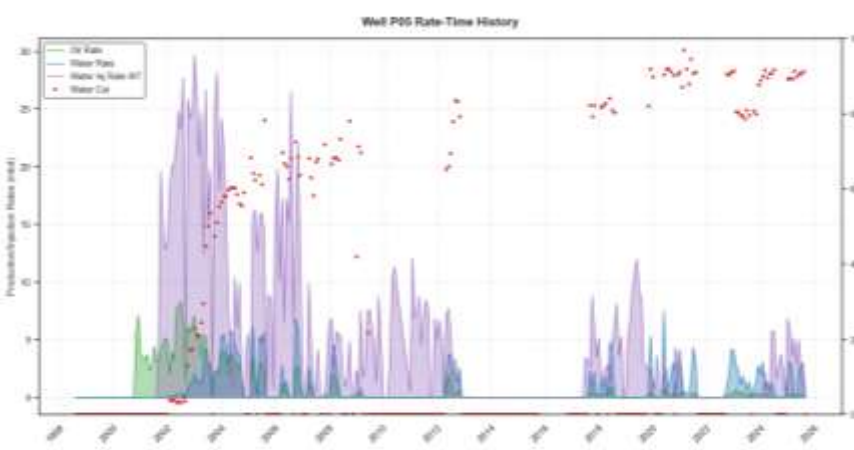
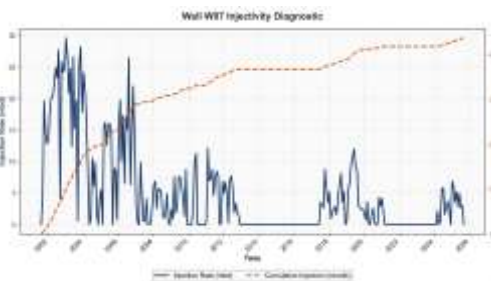


Fig.7: Production and water cut profile for Well P05, paired with individual injectivity diagnostics for supporting injector wells W07

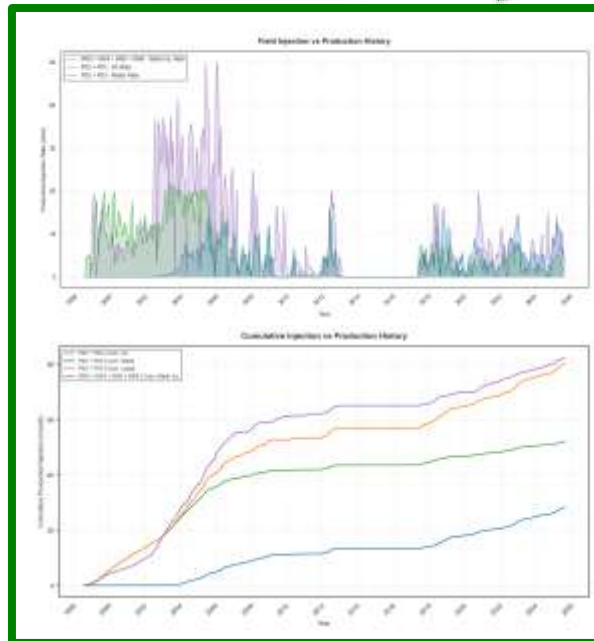


HIGH NET CHANNEL (P02 & P03)

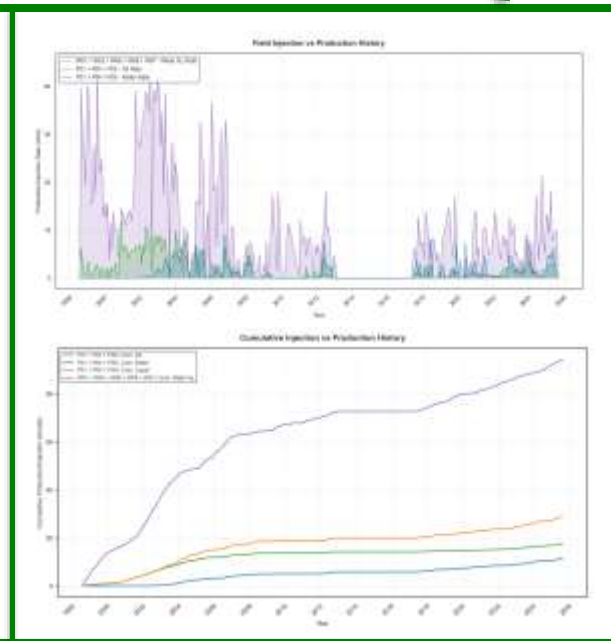
BACKGROUND CHANNEL COMPLEX (P01, P04, P05)



Parameter	Interpretation of P05
Well Performance	Good initial oil → early water breakthrough → later recovery after long shut-in
Injectors Associated	W07
Producer-Injector Relationship	Moderate → early breakthrough via basal high-perm conduit
Pressure Support Pattern	High injection required → later stabilized after long soak
Reservoir Productivity (Spatial)	Located in lobe complex (distal + channel mix) → moderate to high
Lithofacies Control	Lobe sands (lower perm) + basal channel (high perm) → dual flow system
Key Geological Controls	Basal channel acts as fast pathway , lobes act as storage zones

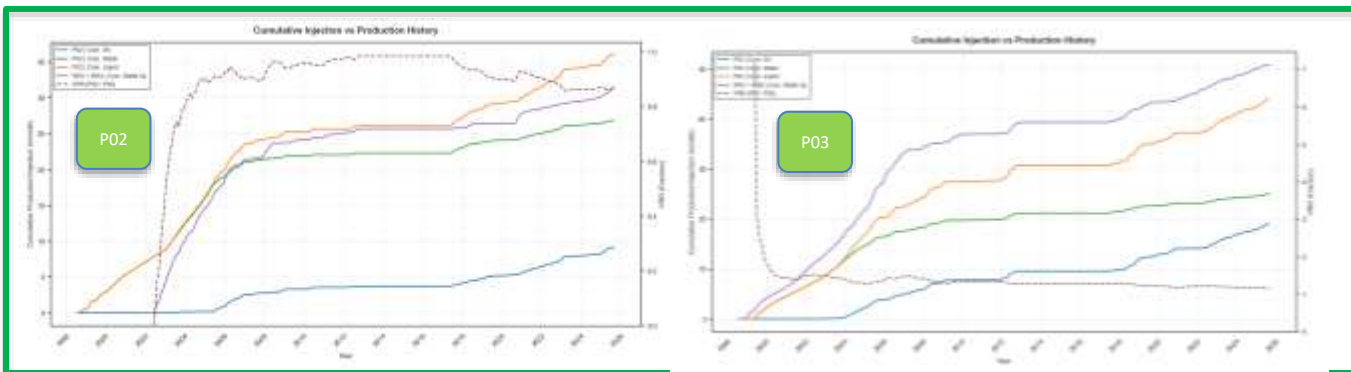


Sustained Deliverability: Steady production plateau → ~22 mbd (2003–2006).
Massive Recovery: ~52 MMSTB cumulative oil → 75% of the entire field's total recovery.
Optimal Sweep: VRR ≈ 1.0 → Perfect pressure support and delayed water breakthrough.



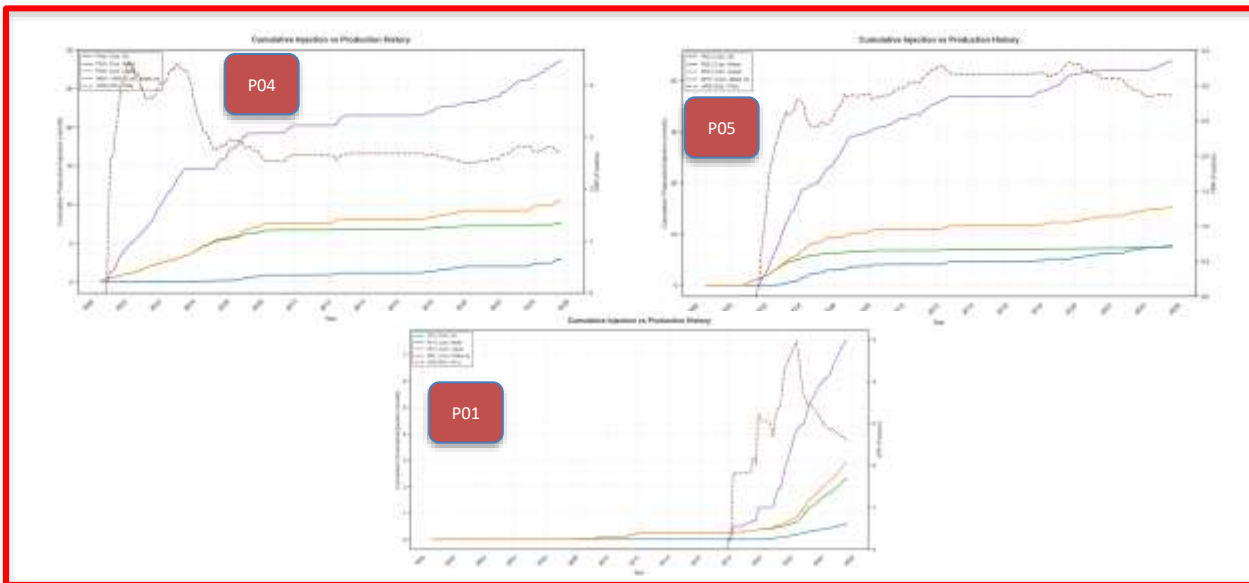
Moderate Deliverability: Combined peak oil rate → ~11 mbd (achieved in mid-2002).
Lower Contribution: ~17 MMSTB cumulative oil → 25% of the entire field's total recovery.
Inefficient Sweep: Massive over-injection (VRR >> 1.0) → Wasted via fault barriers and high-speed channeling.

WATERFLOOD PERFORMANCE



P02 & P03: BALANCED VRR, OPTIMAL PRESSURE SUPPORT IN MAIN CHANNELS

In the primary channel sands (P02, P03), the localized VRR is perfectly balanced (~ 1.0). This indicates highly efficient energy transfer, sustained reservoir pressure, and an optimal, piston-like matrix displacement.



P04 & P05: BACKGROUND MATRIX DEPLETION (GRADUAL WATER CUT RISE DUE TO LOWER OIL AVAILABILITY).

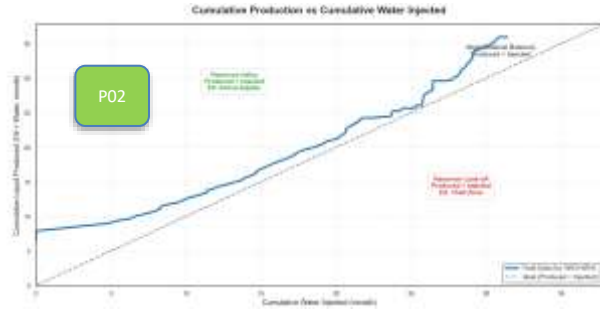
Facies-Driven Depletion: In the tighter background complex (P04, P05), waterflood effectiveness is restricted by rock quality. The gradual, S-curve rise in water cuts indicates natural matrix depletion of smaller oil banks, which required sustained over-injection ($VRR > 1.0$) to force fluid movement.

P01: STRUCTURAL ISOLATION (SEALING FAULT COMPLETELY BLOCKS WATERFLOOD PRESSURE)

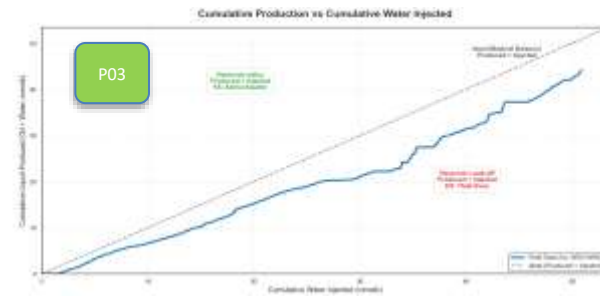
Absolute Structural Barriers: The localized VRR at P01 is entirely disconnected from production. Static mapping confirms a sealing fault is physically blocking the waterflood, leaving virgin oil stranded in an isolated compartment despite heavy offset injection.

WATERFLOOD PERFORMANCE

Water Flood Surveillance Plot

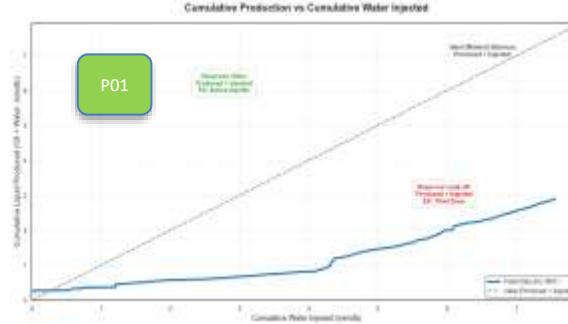
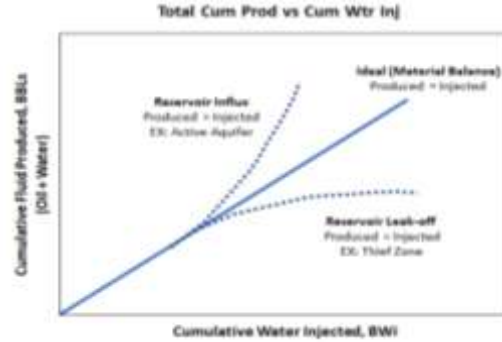


Production ≈ Injection: Balanced voidage (~32 mmstb injected vs. ~36 mmstb produced).
Mechanism: Matrix-dominated displacement demonstrating strong injector-producer connectivity.



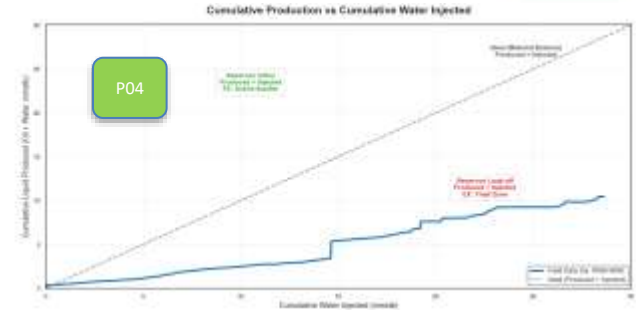
Injection > Production: Moderate over-injection (~51 mmstb injected vs. ~42 mmstb produced).
Mechanism: Moderate fluid leak-off to non-productive zones; while it avoids the severe thief zone channeling seen in underperforming patterns, it still represents wasted injection energy.

HIGH NET CHANNEL

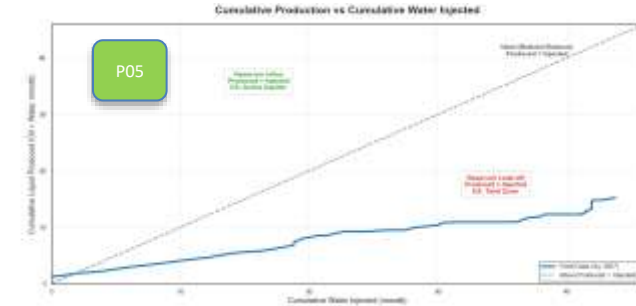


Injection > Production: Severe over-injection (~7.5 mmstb injected vs. only ~3 mmstb produced).
Mechanism: Early fluid leak-off, likely migrating out-of-pattern or into a localized thief zone.

BACKGROUND CHANNEL COMPLEX



Injection >> Production: Extreme over-injection (~29 mmstb inj. vs. ~10 mmstb prod.).
Mechanism: Severe fluid leak-off due to major channeling or a dominant thief zone.



Injection >>> Production: Extreme over-injection (~44 mmstb inj. vs. ~15 mmstb prod.).
Mechanism: Severe leak-off, likely due to out-of-zone injection or a dominant thief zone.

INDICATIONS OF VARIATION IN SWEEP EFFICIENCY OR DISPLACEMENT BEHAVIOR ACROSS DIFFERENT PARTS OF THE RESERVOIR



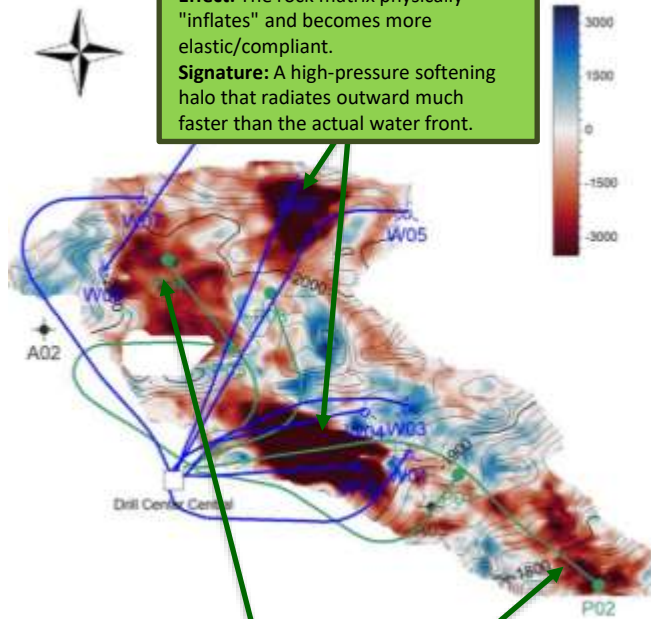
1999-1996

Injectors (W02 & W06): Pore Ballooning

Mechanism: High-pressure water injection (started 1998) increases pore pressure and reduces effective stress.

Effect: The rock matrix physically "inflates" and becomes more elastic/compliant.

Signature: A high-pressure softening halo that radiates outward much faster than the actual water front.



Producers (P02 & P04): Solution Gas Breakout.

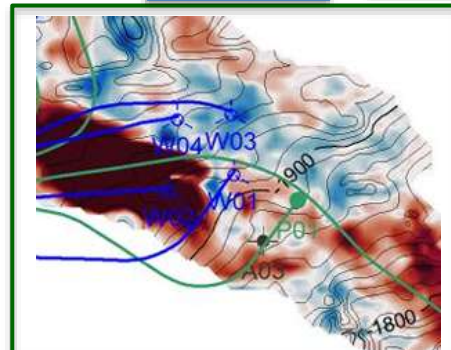
Mechanism: Aggressive fluid drawdown (started 1998) causes local pressure to drop below the bubble point.

Effect: Liberated "free gas" bubbles occupy the pore space.

Signature: Gas is highly compressible, causing a dramatic drop in acoustic velocity and a "bright" softening anomaly.

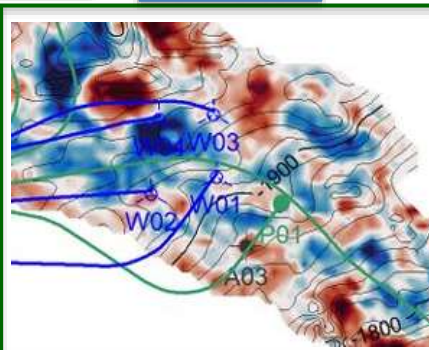


1999-1996



P01- W01+W02

1996-2022



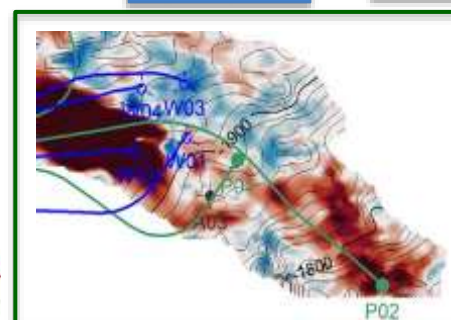
Well P01: Stable Piston Drive (High Conformance)

Extraction Type	Regression Equation (with Time)	R ²
Liquid	$L = -0.379 + (0.097 \times W1) + (0.019 \times W2) + (0.0037 \times T)$	46.2%
Oil	$O = -0.296 + (0.068 \times W1) + (0.015 \times W2) + (0.0029 \times T)$	41.0%

Sweep Efficiency: Excellent Vertical Conformance but Low Volumetric Utilization. 4D seismic shows a unified, consistent blue saturation "bulb" advancing from W01. However, the cumulative plot shows P01 operating in a "Reservoir Leak-off" zone, meaning significant injected volume is lost to the fault/compartiment boundaries rather than reaching the producer.

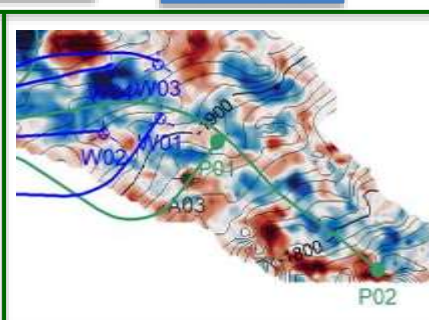
Displacement Behavior: Stable Piston Drive. This is substantiated by the regression model showing a **High Oil-to-Liquid connectivity ratio (~70%)**. The water is effectively pushing a concentrated oil bank ahead of it without significant fingering.

1999-1996



P02- W03+W04

1996-2022



Well P02: Conduit Flow (Channelized Failure)

Extraction Type	Regression Equation (with Time)	R ²
Liquid	$L = 4.778 + (0.396 \times W3) + (0.297 \times W4) - (0.0130 \times T)$	29.0 %
Oil	$O = 4.905 + (0.287 \times W3) + (0.238 \times W4) - (0.0186 \times T)$	45.1 %

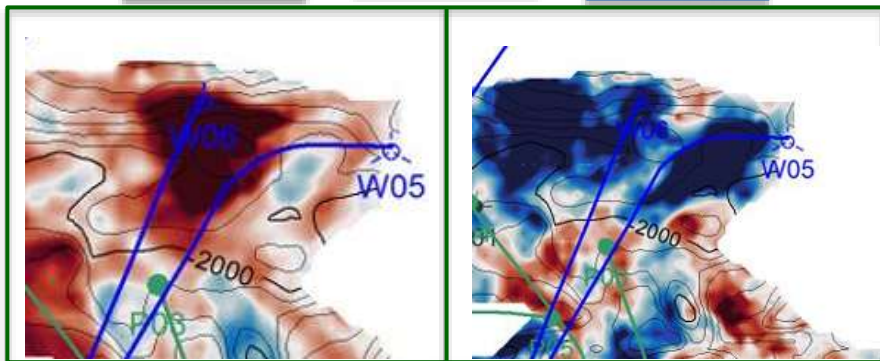
Sweep Efficiency: Poor Vertical and Areal Sweep. Despite being in a "High Net Channel," the cumulative plot shows **Produced > Injected**, indicating an **Active Aquifer Influx**. The 4D seismic shows the blue saturation anomalies appearing as **isolated, narrow streaks** rather than a unified bank. This indicates that the water is not sweeping the full width of the "High Net Channel"

Displacement Behavior: Oil production appears moderately correlated with injection support but liquid production shows much weaker correlation, likely due to **variable water behavior**, suggesting **heterogeneous flow paths** and possible water breakthrough/channeling effects. Further supported by the fact that the **sweep pathway is too long and significantly curvilinear for flow to remain homogeneous.**

1999-1996

P03- W05+W06

1996-2022



Well P03: Gravity-Dominated Recycling (Balanced but Broken)

Extraction Type	Regression Equation (with Time)	R ²
Liquid	$L = 3.235 + (0.556 \times W5) + (0.201 \times W6) - (0.0033 \times T)$	29.2 %
Oil	$O = 3.673 + (0.245 \times W5) + (0.181 \times W6) - (0.0139 \times T)$	44.4 %

4D Seismic: Displays **patchy, non-cohesive lobes**. Water is "thieving" through the highest-permeability stratigraphic layers. Instead of a unified front, the water is selectively "thieving" through isolated high-permeability pathways within the **High Net Channel**.

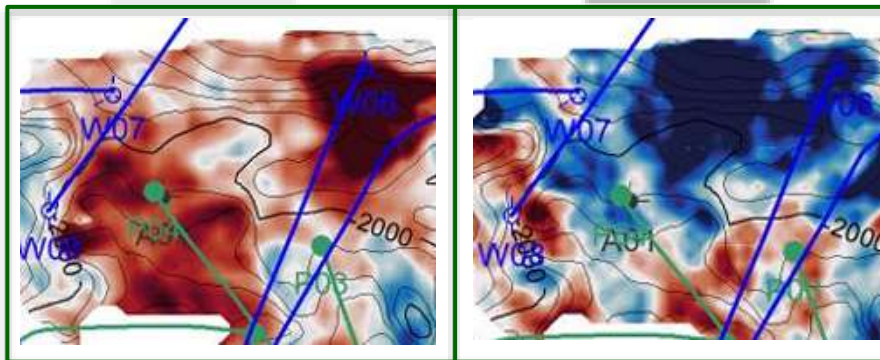
Displacement Behavior: Unstable Water Recycling.

Regression: A low **44% Oil-to-Liquid ratio** (0.245 / 0.556 for W5) indicates the displacement front has collapsed. Most injection energy is now wasted recirculating water through a pre-swept highway at the base of the channel.

1999-1996

P04- W07+W08

1996-2022



Well P04: Cumulative Leak-off

Extraction Type	Regression Equation (with Time)	R ²
Liquid	$L = 1.560 + (0.078 \times W6) - (0.227 \times W8) - (0.0045 \times T)$	14.6 %
Oil	$O = 1.667 + (0.056 \times W6) + (0.209 \times W8) - (0.0087 \times T)$	36.2 %

Sweep Efficiency: Poor (Localized/Bypassed). Seismic shows water lobes diverted away from the producer. Operating in a severe **Leak-off zone**, P4 is deprived of injection support as fluids migrate into external thief zones or faults.

Displacement Behavior: Cumulative Leak-off confirm injection is thieved by fault systems. The water bank is diverted away from the producer, preventing a cohesive pressure front. Disconnected pressure drive caused displacement to collapse into **viscous fingering** supported by low liquid correlation indicating variable liquid behavior. Erratic **80% water-cut spikes** indicate "short-circuiting" through high-perm micro-streaks rather than a solid oil bank push.

Induced Fracturing Due to Over-Injection (Operational)

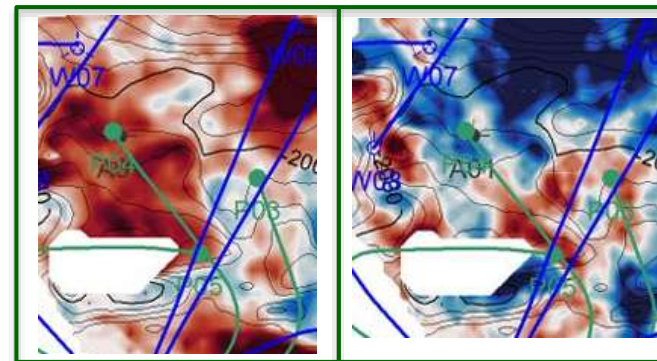
Excessive injection pressure likely induced fractures, creating a high-permeability conduit that short-circuited the waterflood to P04, as supported by the water-cut trend.

Well P05: Mature Water Recycling

1999-1996

P05- W07

1996-2022



Sweep Efficiency: Very Poor (Volumetric Failure).

4D Seismic: Minimal saturation change is observed around P05 compared to the northern wells. Water from W07 is almost entirely diverted North/Northwest, leaving P05 in a "shadow zone."

Volumetrics: Operating in a severe **Leak-off regime**. Nearly 45 mmstb injected at W07 has yielded only ~15 mmstb of total liquid at P05, proving a massive volumetric disconnect.

Extraction Type	Regression Equation (with Time)	R ²
Liquid	$L = 0.956 + (0.210 \times W7) - (0.0020 \times T)$	36.1 %
Oil	$O = 0.907 + (0.131 \times W7) - (0.0047 \times T)$	48.4 %

Displacement Behavior: Mature Water Recycling.

Connectivity: Regression shows moderate liquid support (0.210 W7) but very low oil coefficient (0.131 W7), resulting in a **62% displacement efficiency**.

Production Signature: Rapid breakthrough and high stable water-cut (~90%) suggest that P05 is merely recirculating water through a thin, fully-swept basal layer while bypassing attic oil.

RESERVOIR RISKS AND UNCERTAINTY

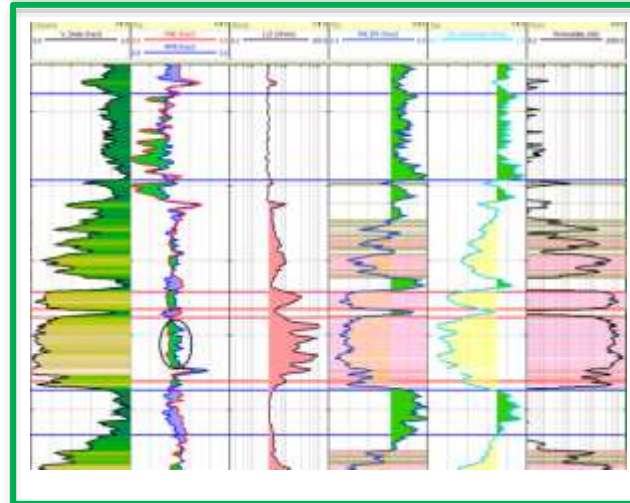
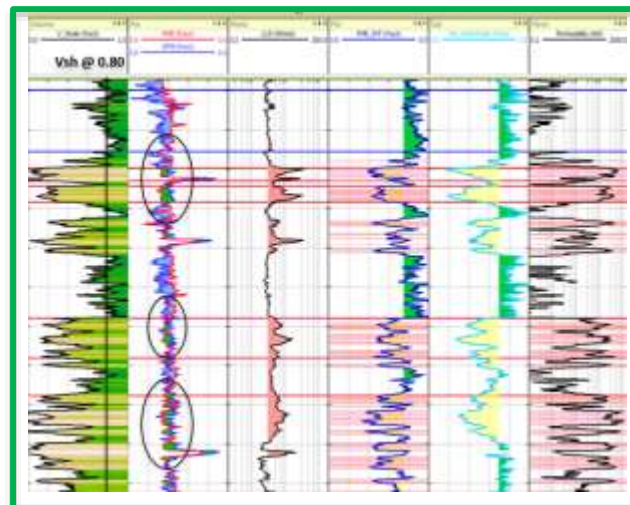
LITHOFACIES & CONNECTIVITY (STATIC)

Well Group	Log Analog	Facies Association	Deliverability Impact	Recovery Risk
P02, P03	A03 (Blocky)	Axial High Net Channel (PF1)	Highest: Massive, clean sand allows for maximum flow rates and high RQI (0.75).	Short-Circuiting: No vertical baffles to stop "thief zones." Water rushes through the entire block (e.g., P02 water spike).
P01, P04	A01 (Serrated)	Background Channel / Margin (PF2/3)	Low to Moderate: Shale "teeth" create tortuous flow paths, choking initial rates.	Bypassed Oil: Vertical sweep is uneven. Water sweeps individual "teeth" but leaves oil trapped behind shale drapes.
P05	Mixed/Lobe	Lobe Complex (PF3 Sand-prone)	Inconsistent: Dependent on thin sand sheet connectivity.	Compartmentalization: High risk of being disconnected from pressure support.

The Axial Argument: P02 and P03 are located in the "sweet spot" of the high-amplitude seismic fairway. Since **A03** shows a "Blocky" log with high Rock Quality (RQI 0.75), it represents the **Amalgamated Channel Axis**. These production wells show the high rates and rapid water spikes that only occur in clean, blocky sands without vertical baffles.

The Marginal Argument: P01 and P04 are positioned toward the edges of the channel fairway or in the "Background Complex."
A01 is the perfect analog because its "**Serrated**" log (50% Shale Volume) explains the high flow resistance and "baffled" pressure support observed in these wells. They are physically located in the heterolithic overbank/fringe areas.

The Lobe Argument: P05 is situated in a **Lobe Complex** (PF3) on the down-dip flank. It is associated with the **fringe of A01/A03** because it has the high porosity of the lobes but the connectivity risks of a distal setting. Its "Injection Slave" behavior proves it is in a sand-prone but laterally restricted package.



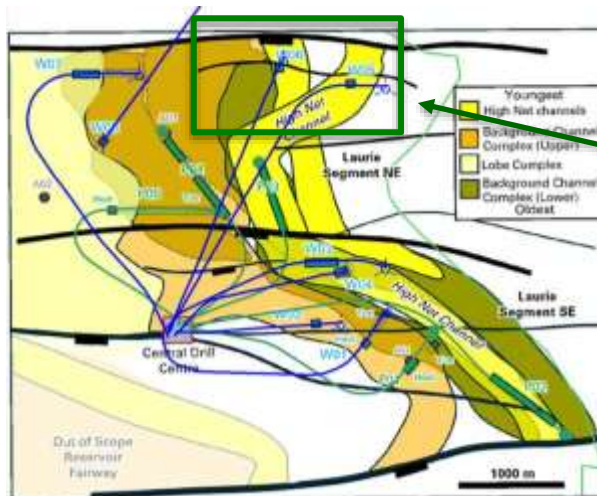
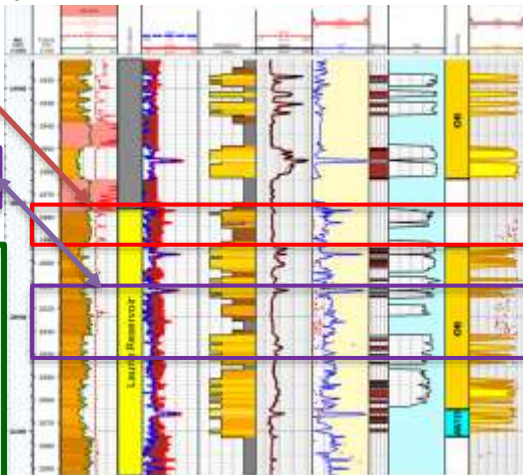
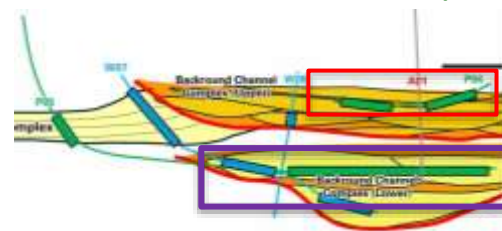
A1

Net-to-Gross (%)	36.46
Avg Vsh	0.50
Avg Φ_{eff}	0.17
Avg Sw	0.64
Avg Permeability (mD)	77.34
Flow Capacity (kh)	455.02
Rock Quality Index (RQI)	0.35

A3

Net-to-Gross (%)	35.12
Avg Vsh	0.55
Avg Φ_{eff}	0.15
Avg Sw	0.61
Avg Permeability (mD)	123.95
Flow Capacity (kh)	2267.58
Rock Quality Index (RQI)	0.35

STRUCTURALLY CONTROLLED (STATIC)



P4 and P5 are in thin Lobe Complexes (PF3). Unlike thick axial sands, these thin layers are easily "punctured" by faults.
Risk: Injected water follows the fault planes vertically into non-productive deeper zones instead of moving horizontally to sweep oil.

Target Facies & Lithological Risk

Facies Mismatch: P5 and P4 horizontal wells are landing in **Amalgamated Sand Channels** and **Heterolithic Sands**, but logs reveal a dominance of **Mud Streaks**.

Lean Oil Characteristics: Production confirms a **Lean Oil Reservoir**—not due to lack of oil, but because it is trapped in thin **intercalations and patches**.

Depth Correlation: TVD tracking against **Well A1** data confirms the laterals are predominantly skimming the **mudline**, leading to suboptimal reservoir contact.

Petrophysical Impact on Production

Low Resistivity Signal: Continuous **low-resistivity zones** indicate the wells are tapping directly into **mud layers** rather than high-permeability sand.

Poor Deliverability: The presence of **shale/mud streaks** creates vertical flow barriers, resulting in poor primary production and restricted drainage.

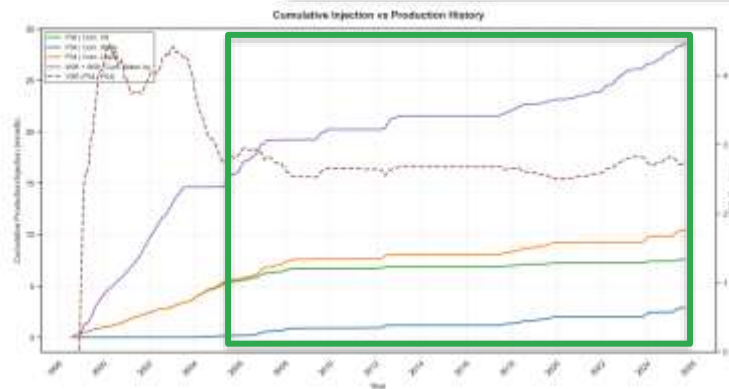
Injection Failure & Water Management

Short-Circuiting: Waterflooding is failing to achieve "piston-like" displacement; injected water is bypassing the oil-bearing matrix.

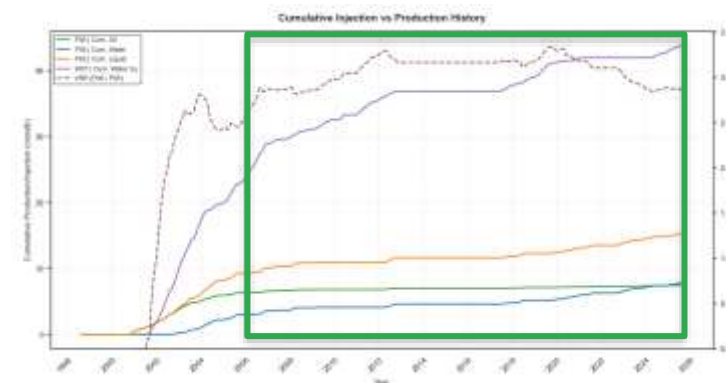
"Infinite Water" Production: Water is channeling through **faulted zones and connected mud streaks**, resulting in immediate breakthrough at the producer.

Congeaed Risk: The reservoir architecture allows water to find the path of least resistance, leaving the **oil-bearing patches** unswept and stranded.

EVIDENCE : THE MASSIVE "VOIDAGE GAP" (CUMULATIVE INJECTION >> CUMULATIVE PRODUCTION)

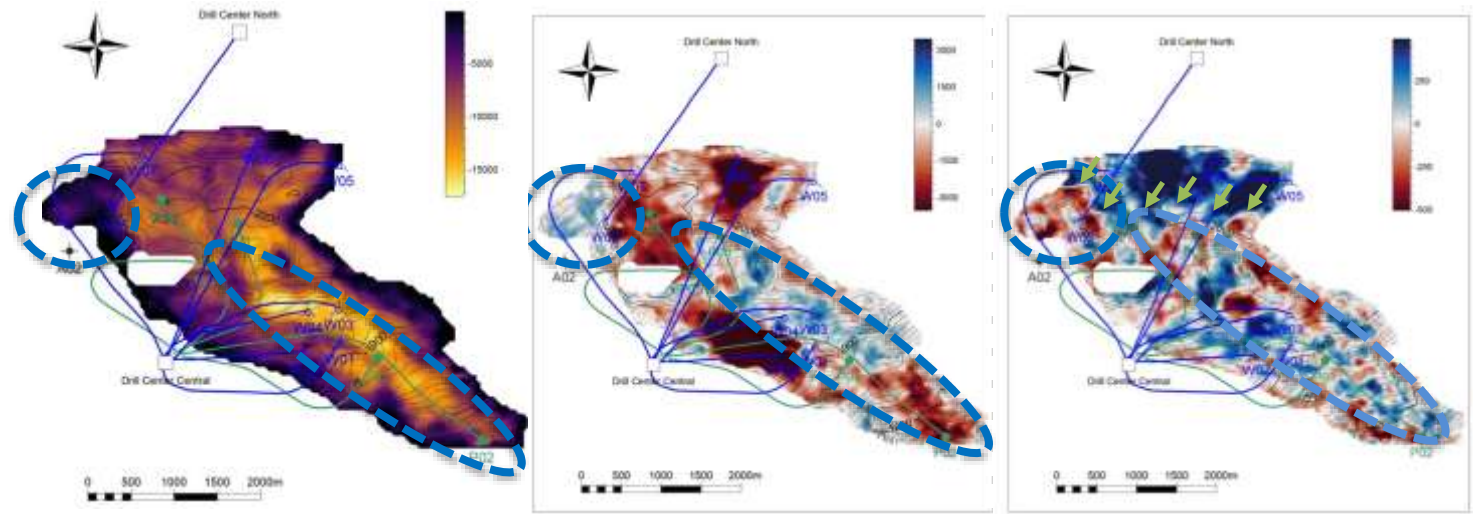


P04: Extreme over-injection (~29 mmstb inj. vs. ~11 mmstb prod.).



P05: : Extreme over-injection (~44 mmstb inj. vs. ~16 mmstb prod.).

DYNAMIC DATA REFINE UNDERSTANDING OF THE RISKS AND UNCERTAINTIES



OBSERVATION 1	INITIAL STATIC UNDERSTANDING (SNA)	4D DYNAMIC REFINEMENT (TIME-LAPSE)	NEW RESERVOIR REALITY
A02 Target Area	No significant SNA evidence; high Saturation Uncertainty .	4D reveals a coherent Migration Front moving West.	Target is being actively charged with oil .
Inter-well Connectivity	Assumed poor lateral communication/isolation.	Sweep Pattern proves efficient hydraulic connectivity from central injectors.	Confirmed Waterflood Push toward the appraisal zone.
Strategic Status	Categorized as a high-risk Frontier/Step-out .	Proved to be a high-confidence Appraisal Target .	De-risked by replacing static doubt with Proven Sweep Evidence .

OBSERVATION 2	INITIAL STATIC UNDERSTANDING (SNA)	4D DYNAMIC REFINEMENT (TIME-LAPSE)	NEW RESERVOIR REALITY
High-Net Channels	High SNA mapped as the primary HC Fairway.	Blue Anomalies confirm extensive Water-for-Oil replacement.	High-quality zones are now swept/watered-out .
Peripheral Zones	Low SNA flagged as low-potential/non-productive.	Red Anomalies identify unswept, pressure-supported Oil Banks .	Flanks hold significant Bypassed Pay potential.
Saturation Risk	Assumed rock quality = oil presence.	Proved Rock Quality ≠ Current Saturation .	High-k channels act as "thief zones" for water.

STOIIP STATIC	STOIIP DYNAMIC
Not considered earlier	Approx. 12.9 mmstb

RESERVOIR SURVEILLANCE STRATEGY

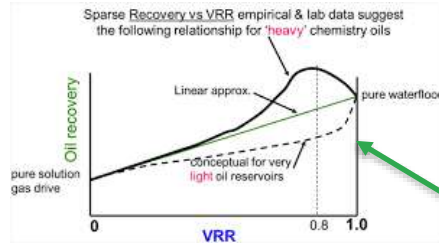
TRANSITIONING FROM REACTIVE MONITORING TO PROACTIVE RESERVOIR MANAGEMENT TO MAXIMIZE ULTIMATE ECONOMIC RECOVERY.

PRIMARY OBJECTIVES

Maximize Volumetric Sweep: Ensure uniform areal and vertical displacement across heterogeneous facies (e.g., intercalated sands).

Manage Reservoir Energy: Maintain a balanced Voidage Replacement Ratio (VRR ~1.0) to provide pressure support without exceeding rock fracture gradients.

Diagnose & Mitigate Conformance Issues: Rapidly identify high-permeability thief zones, fault diversion, and operationally induced channeling.



FIELD-SCALE (MACRO ENERGY)

Continuously track **Field VRR and Q vs. Np**.

Rapid decline or VRR < 1 triggers injection increases; VRR > 1 triggers channeling investigations.

SECTOR-SCALE (REGIONAL SWEEP)

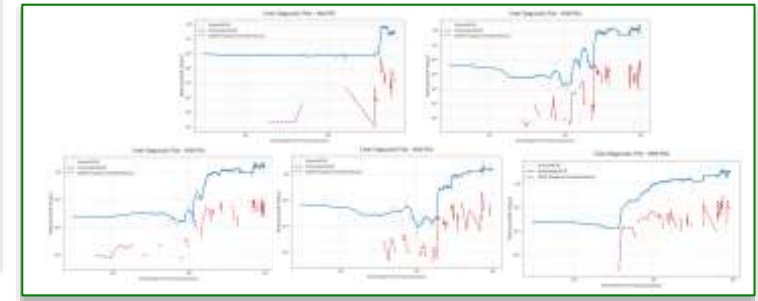
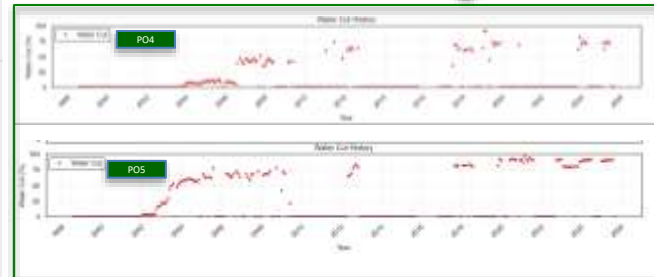
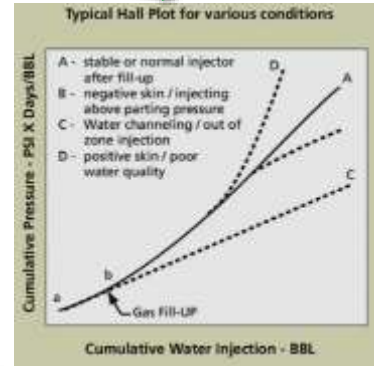
Analyze **Water Cut vs. Time curves**. Smooth S-curves confirm matrix sweep; erratic/steep rises indicate severe heterogeneity, prompting conformance control.

PATTERN/WELL-SCALE (CONNECTIVITY & INJECTIVITY)

Utilize continuous **Pressure Monitoring for Hall plots** on injectors to rapidly identify fracturing (decreasing slope) or formation damage/plugging (increasing slope)

Execute Step-Rate Tests (SRTs): Gradually increasing injection rates while monitoring pressure to determine the Formation Parting Pressure (fracture gradient) of different rock types.

Utilize **Chan Plots (WOR)** and Time Lag Analysis to diagnose breakthrough mechanisms



RESERVOIR SURVEILLANCE STRATEGY

TARGETED DATA ACQUISITION & MONITORING TECHNIQUES HEADLINE: DEPLOYING ADVANCED SURVEILLANCE TO RESOLVE SPECIFIC WELL-LEVEL DYNAMIC UNCERTAINTIES.



P01 (Delayed Response & Fault Compartmentalization):

Techniques: Pressure Transient Analysis (PTA / Interference Tests) and 4D Seismic.

Application: PTA will define structural boundaries and mathematically prove if the nearby fault is a sealing barrier or a leaky conduit. 4D seismic will visually confirm if injected fluids are being diverted away from P01 into the fault system.

P02 (Injection Sensitivity & Early Breakthrough):

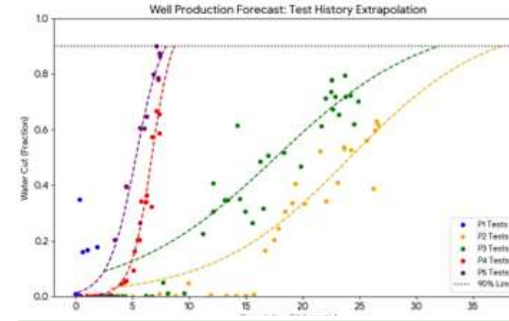
Techniques: High-frequency Production/Injection Data (Rate, VRR) and BHP Monitoring.

Application: Build a localized, dynamic VRR model to optimize the injection strategy. By tracking WC sensitivity to injection pulses, we can establish the exact threshold rate that provides pressure support without triggering early water breakthrough, securing the 11 MMSTB remaining potential.

P03 (Injector Contribution & Channeling):

Techniques: Interwell Chemical Tracers.

Application: Injecting unique tracers into W05 and W06 to definitively allocate the volumetric contribution of each injector to P03. Measuring transit times will quantify early breakthrough paths and confirm if the 7 MMSTB 10-year target is viable under re-balanced injection.



Logistic Fit of test history data set:- Water cut and cumulative oil production

P04 (Marginal Well Evaluation & Abandonment):

Techniques: 4D Seismic integrated with Reservoir Simulation & History Matching, the Thomas-Stieber program, and seismic attributes.

Application: Utilize the Thomas-Stieber program and seismic attributes for precise sand mapping to identify remaining sweet spots in this highly heterogeneous region. The dynamic simulation, matched against 25 years of history, will conclusively determine if the 1.2 MMSTB remaining reserves justify continued OPEX or if well abandonment is the optimal decision.

P05 (High WC & Recompletion Potential):

Techniques: Production Logging Tools (PLT), high-resolution Image Logs, and Cased Hole Saturation Logging (e.g., PNL/RST).

Application: Use image logs for high-resolution facies classification of the intercalated sequences. Run PLTs to precisely identify which shale/sand intervals are dominating the 85% water influx. Execute Cased-Hole Saturation Logging (PNL/RST) by logging mature producers behind the casing to measure current water saturation and identify bypassed vertical pay zones containing virgin oil, justifying mechanical recompletion of the 11 MMSTB remaining potential.

RESERVOIR SURVEILLANCE STRATEGY

UNCERTAINTY REDUCTION & DECISION SUPPORT:
TRANSFORMING DYNAMIC SURVEILLANCE DATA INTO
ACTIONABLE FIELD DEVELOPMENT DECISIONS.

INJECTION OPTIMIZATION (P02 & FIELD VRR)

Resolves the **uncertainty between matrix flow and channeling**. Oil sweep towards the banks of background channel flank.
Decision: Reconfigure injector-producer patterns and implement **cyclic/pulsed injection to maximize matrix imbibition and flatten the water cut trend**.

CONFORMANCE CONTROL & WORKOVERS (P05)

Resolves **uncertainty regarding vertical sweep and water entry points**.
Decision: Execute targeted workovers using mechanical plugs or polymer gels to isolate water-bearing shales, followed by recompletion in virgin oil sands identified by RST logs.

INFILL DRILLING & PATTERN REALIGNMENT (P01 & P03)

Resolves **structural connectivity and fault diversion risks**.
Decision: If tracers and PTA prove W05/W06 are short-circuiting, or P01 is completely isolated by the fault, **capital will be reallocated to drill new infill wells in the unswept "shadow zones"** identified on sector bubble maps.

ECONOMIC END-OF- LIFE DECISIONS (P04)

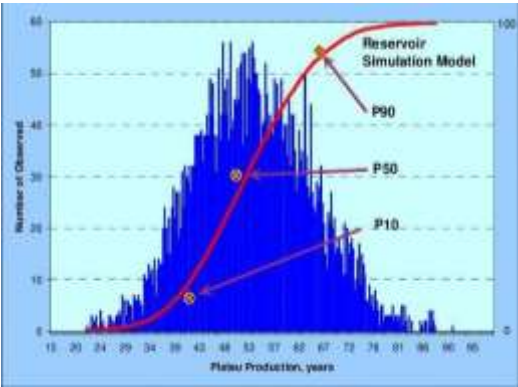
Removes the guesswork from mature asset management.
Decision: Confirmed low hydrocarbon saturation via integrated 4D/simulation leads to immediate, **justified well abandonment**, saving lifting costs. Monitor the influence of newly drilled W08 injection well in production and decide whether to operate production on this well.

MONTE CARLO SIMULATION INTEGRATION

Uncertainty Source	Input Method / Variable	Specific MCS Output Value	Value Add & Meaning
Mud Streaks / Shale Barriers	Stochastic Modeling of Kh/Kv ratios across heterolithic facies.	P90 (Low Case) Volumetrics	Defines the "conservative" remaining oil if vertical barriers are continuous.
Channeling / Thief Zones	Sensitivity Analysis on Sweep Efficiency and NTG.	Tornado Plot & P50 (Mean)	Identifies the primary "driver" of water breakthrough; defines the most likely recovery.
4D Seismic Migration	Probabilistic Timing of saturation front velocity.	85% Confidence Interval (CI)	Provides a numerical Probability of Success (PoS) for production from Well A02.
Water Cut Steepness	Distribution Curve of sigmoid decline rates.	Cumulative Frequency Histogram	Predicts the probability of Economic Truncation (e.g., a 70% chance Well P5 fails in <1 year).

Monte Carlo Simulation Integration:
Rather than relying on single-point deterministic estimates for the decisions above, we will embed **Monte Carlo simulation** directly into our strategy plan.

Probabilistic Financial Modeling: By feeding the uncertainty ranges from our surveillance data (e.g., fault volume ranges from PTA, or remaining oil saturations from RSTs) into the simulation engine, we can calculate comprehensive, risk-weighted financial metrics for every proposed intervention



Due to insufficient data availability, we were unable to perform the Monte Carlo simulation for the reservoir.

DEVELOPMENT STRATEGY AND OPPORTUNITY EVALUATION



To maximize recovery and exploit remaining development opportunities, the following multi-phased approach is proposed. This concept focuses on reconfiguring the current well patterns to target unswept oil identified through 4D seismic analysis and well log petrophysics.

TARGET: UNSWEPT ATTIC OIL IDENTIFIED BY A PROMINENT 4D NEGATIVE AMPLITUDE ANOMALY (RED ZONE).

NEW INFILL INJECTOR W10

W10 drilled in lobe complex will work in tandem with the existing **W07** injector to create a **Peripheral Line-Drive** configuration. W10 gives Primary southern drive; provides dedicated sweep to A02. **Well W07 (Existing)** provides Northern flank support; prevents attic oil escape.

A02 APPRAISAL → PRODUCTION WELL (P08)

Reservoir Quality: Well A02 logs confirm high **Net-to-Gross (NTG)** with blocky sand signatures and a distinct **Density-Neutron crossover**, indicating a significant untapped hydrocarbon column.

DEVELOPMENT CONCEPTS
Infill Drilling & Pattern
Realignment.

A01 APPRAISAL → INJECTION WELL

The **serrated log signature** of A01 confirms poor reservoir quality (high shale content), making it an ideal candidate for injection well conversion rather than production well.

P05: Targeting High-Saturation "Sweet Spots"

Objective: To align the production interval with peak reservoir quality identified in petrophysical logs.
Mechanism: By extending perforations further into the adjacent heterolithic facies, the P03 drainage radius is moved into closer proximity to the A01 injection front. This reduces the distance for fluid travel across the **basal scour**, ensuring that the injected water effectively sweeps the distal channel edges toward the P03 wellbore.

PRODUCTION DECLINE IN P04 AND P05

Both **P04** and **P05** are exhibiting a progressive **decline in production rates**, likely due to reservoir pressure depletion and the lack of localized voidage replacement. To arrest this decline, the field development plan must pivot toward enhanced pressure support.

DEVELOPMENT CONCEPTS: Well
Conversion &
Cross-Facies
Pressure
Maintenance.

P03: Strategic Extension toward Heterolithic Facies

Objective: To enhance the pressure-link with the newly converted **A01** injector.
Mechanism: By extending perforations further into the adjacent heterolithic facies, the P03 drainage radius is moved into closer proximity to the A01 injection front. This reduces the distance for fluid travel across the **basal scour**, ensuring that the injected water effectively sweeps the distal channel edges toward the P03 wellbore.

Facial Connectivity (P04)	Erosional Surface Enhancement (P03):	Lower Facial Connectivity (P05)
A01 and P04 reside within the same channel facies, ensuring direct pressure support and high sweep efficiency for the P04 drainage cell.	Although A01 and P03 are in different facies, the Basal Scour acts as a major erosional unconformity. This surface enhances vertical and lateral permeability, allowing the water-drive from the A01 injector to "cross-flow" into the adjacent facies and support the P03 producer.	The P05 producer was strategically drilled to target the high-quality High Net Sands at depth. Petrophysical correlation confirms that the A01 wellbore shares this identical facies. This shared lithology establishes a high-degree of reservoir connectivity between the two wells.

DEVELOPMENT STRATEGY AND OPPORTUNITY EVALUATION



TARGETING FAULT-BOUNDED BYPASSED OIL: STRATEGIC SEGMENT INFILL

The presence of a persistent "**Red Zone**" (**unaffected acoustic impedance**) in 4D seismic suggests a significant volume of unswept hydrocarbons. Despite being surrounded by injectors, this zone remains untapped, indicating a **stratigraphic or structural baffle** that isolates the reservoir from the current flood front.

GEOLOGICAL DRIVERS:DUAL-NATURE FAULTING

The **Major Fault** in the region appears to be a **conduit**, allowing hydrocarbons to migrate along a specific pathway but potentially bypassing the adjacent compartment.

The **Minor Faults** act as **sealing barriers**, compartmentalizing the reservoir and preventing current injectors from sweeping this specific zone.

MITIGATING BYPASS AND FACIES MIGRATION: P02 OPTIMIZATION STRATEGY

Time-lapse **4D seismic** confirms this theory; the **Red Zones** (unswept/high-saturation oil) are visibly shifting from the core High Net Channel into the adjacent **Heterolithic Complex**.

PRODUCTION TREND

Rate-Time History for P02 shows a significant long-term decline from its initial peak. Continued extraction from the current wellbore risks total pressure depletion without capturing the volumes that have moved laterally.

"HIGH-PERM" PROBLEM

While the High Net Channel offers excellent porosity and permeability, these properties combined with the connectivity provided by the **basal scour** have allowed hydrocarbons to migrate rapidly.

DEVELOPMENT CONCEPTS: Lateral Infill & Secondary Facies Drainage.

Producer P09 within the adjacent heterolithic facies. This well is strategically positioned to capture the hydrocarbons being pushed by the natural and induced pressure drive from the High Net Channel.
Pressure Management: By introducing P09, we reduce the burden on P02 and prevent the total loss of reservoir potential. This allows for a more controlled, "dual-point" drainage of the Laurie Segment SE.

Appraisal Validation: Utilize the Phase 1 proposed appraisal well location (coinciding with P7) to confirm reservoir quality and fluid saturation

Production Deployment: Upon positive appraisal results (good hydrocarbon signatures), drill/complete production wells **P7** and **P6** to directly drain this isolated compartment.

Strategic Injection (W09): * Introduce a new injector, **W10**, to provide dedicated pressure support within this fault-bounded block. This creates a localized sweep mechanism independent of the existing, bypassed flood front.

DEVELOPMENT CONCEPTS Phased Appraisal-to-Production & Structural De-risking

A02 (P08) Attic Oil Capture

Resources: High-volume potential in thick blocky sands (~20m net pay) validated by 4D seismic "Red Zone" anomalies.
Profile: High-rate flush production with a steady plateau supported by a peripheral line-drive injection cluster (W07/W10).

P03/P04/P05 Revitalization (A01 Conversion)

Resources: Incremental recovery through direct shared-facies connectivity and basal scour cross-flow.
Profile: Arrests the annual production decline, creating a secondary "hump" in the profile through restored pressure support and voidage replacement.

Strategic Segment Infill (P07, P06, W09)

Resources: Bypassed volumes isolated by sealing minor faults; de-risked via appraisal-to-production phasing at the P06 location.
Profile: Moderate initial potential from untapped compartments, requiring dedicated W09 injection to maintain pressure against structural baffles.

P09 Migrated Resource Capture, with W11 Injection

Resources: Recovery of the "oil bank" migrated from high-perm channels into the adjacent Heterolithic Complex via the basal scour.
Profile: Stable, long-term "tail" production; flow rates are lower than channel sands but capture volumes otherwise lost to facies migration

Model	STOIIP (MMBO)
Max	57.06
Most Likely	21.8
Min	4.97

Stage 1 (Static Volumetrics): STOIIP was calculated using a probabilistic approach (Min/Most Likely/Max). The **Most Likely STOIIP of 21.8 MMBO** serves as the baseline for our "remaining oil" calculations.

Validation of Results (Static vs. Dynamic)
The critical finding of this study is the **convergence of data**. The dynamic extrapolation of current well performance aligns closely with the remaining oil volumes calculated from the static model. This alignment confirms that the reservoir is performing predictably according to the identified sand bodies and facies distribution, significantly de-risking the proposed infill targets (P06, P07, P08, and P09).

Stable Producers (P2, P3): These wells form the backbone of the field's base production, showing long-tail profiles.

Declining/End-of-Life Assets (P4, P5): P5 is approaching the 90% water cut limit (<1 year remaining life). This validates our strategy to pivot from these wells toward the **A01 conversion** and **A02 attic oil capture** to replace lost capacity.

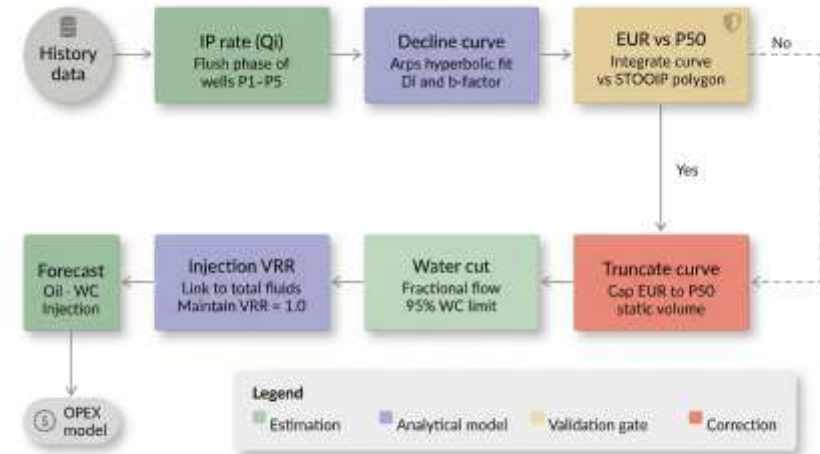
Infill/New Capacity (P06, P07, P08, P09): These profiles are expected to mimic the early-life performance of P2, capturing the "Red Zone" saturations that existing wells cannot reach.

Stage 2 (Dynamic Calibration): We performed **Test History Extrapolation** by plotting Water Cut vs. Cumulative Oil (image_4ab25e). By extrapolating these trends to a **90% Water Cut limit**, we derived the **Ultimate Recovery (EUR)** for each existing well.

KEY UNCERTAINTIES:-

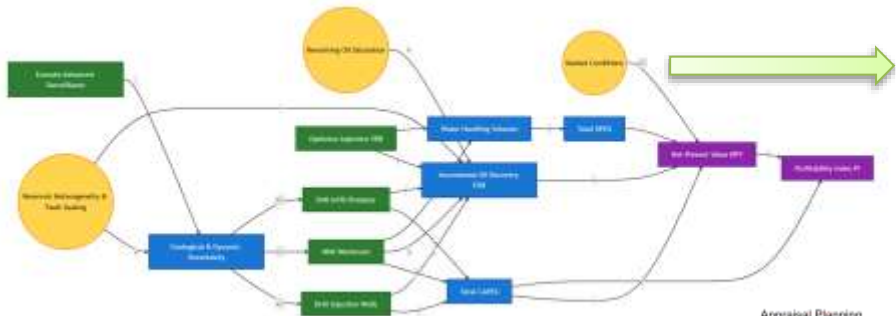
- Non-Linear Breakthrough:** Regression models assume smooth water-cut rises, often failing to predict "jumps" caused by high-perm conduits like the **basal scour**.
- Data Sparsity Bias:** Limited well data risks **overfitting historical trends**, potentially ignoring the positive impact of new pressure support (e.g., A01 conversion).
- Structural Baffles:** 4D "Red Zones" may overstate volumes if **sub-seismic sealing faults** or heterolithic facies compartmentalize the reservoir

METHOD TO ESTIMATE POTENTIAL PRODUCTION PERFORMANCE.



Well ID	Current Cum. Oil (Np, mmstb)	Ultimate Recovery @ 90% WC (mmstb)	Remaining Oil (mmstb)	Remaining Life (Years)
P1	2.32	—	—	—
P2	26.84	37.50	10.66	~40.0 Years
P3	25.06	32.02	6.96	~8.2 Years
P4	7.57	8.72	1.14	~6.8 Years
P5	7.46	7.99	0.53	< 1 Year

DEVELOPMENT STRATEGY AND OPPORTUNITY EVALUATION



FIELD DEVELOPMENT INFLUENCE CHART

Maximizing field value requires reducing uncertainty, optimizing recovery, and controlling costs under market conditions.

Flow: Uncertainty → Decisions → Outcomes → Value

■ Decisions
● Uncertainty
■ Outcomes
■ Value
 Impact:
 (+) increase
 (-) decrease
 (±) variable

FIELD DEVELOPMENT DECISION TREE

Objective: Maximize ENPV using a probabilistic approach to target reservoir heterogeneity, maintaining optimal injection (VRR ~1.0) to control OPEX.

Total Value: \$2.55 Billion (Most likely estimated Undiscounted Net Cash Flow).

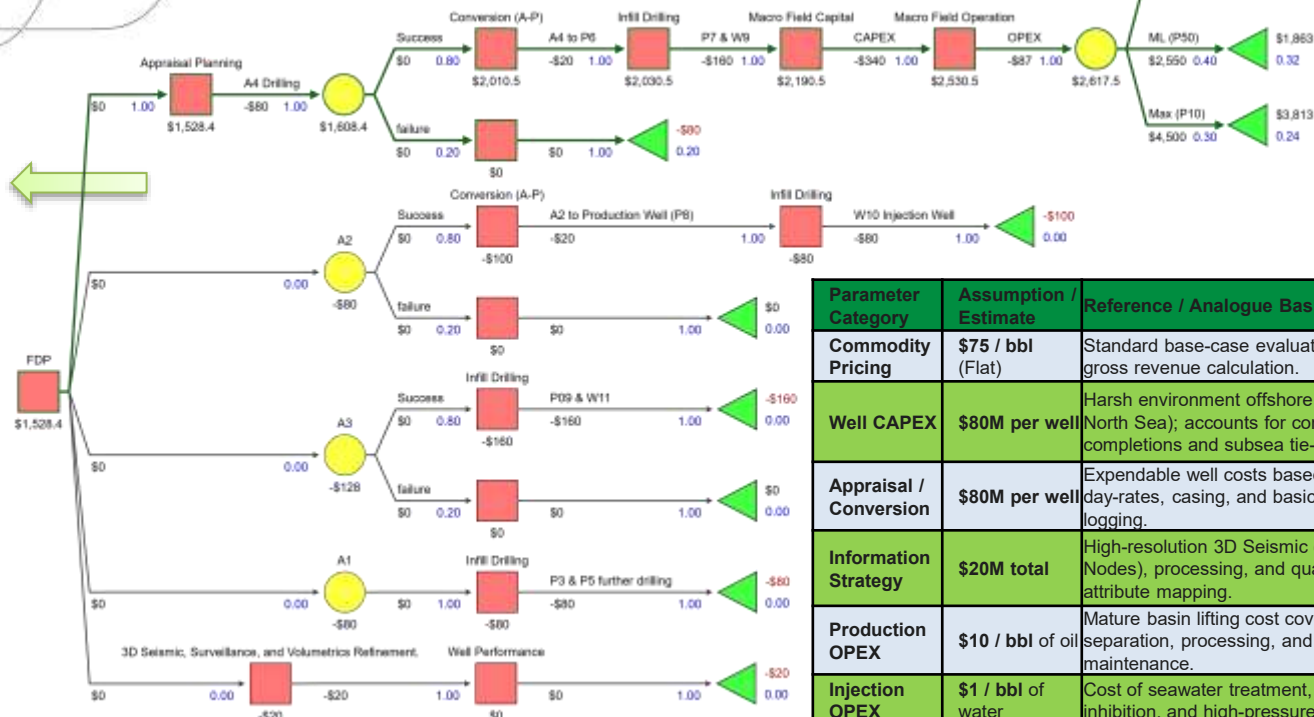
Phased Execution:

- **De-Risk:** Invest \$20M in 3D Seismic to resolve geological uncertainty, specifically targeting the unswept stratigraphic baffle in High net channel sand.
- **Core Development:** Execute infill drilling to increase sweep within the primary anchor channels (Group 6, \$660M NCF).
- **Expand Sweep:** Capture bypassed oil in the Background channel complex in the southern zone and optimize sweep in the NW lobe complex flank.
- **Manage Risk:** Suspend operations on marginal compartments to protect topside fluid-handling capacity and mitigate downside price volatility.

■ Decision Node

● Chance Node

▲ Terminal Node



Parameter Category	Assumption / Estimate	Reference / Analogue Basis
Commodity Pricing	\$75 / bbl (Flat)	Standard base-case evaluation pricing for gross revenue calculation.
Well CAPEX	\$80M per well	Harsh environment offshore analogue (e.g., North Sea); accounts for complex completions and subsea tie-backs.
Appraisal / Conversion	\$80M per well	Expendable well costs based purely on rig day-rates, casing, and basic wireline logging.
Information Strategy	\$20M total	High-resolution 3D Seismic (Ocean Bottom Nodes), processing, and quantitative attribute mapping.
Production OPEX	\$10 / bbl of oil	Mature basin lifting cost covering separation, processing, and platform maintenance.
Injection OPEX	\$1 / bbl of water	Cost of seawater treatment, chemical inhibition, and high-pressure injection.
Uncertainty Weights	30% / 40% / 30%	Swanson's Rule applied to P90 (Min), P50 (Most Likely), and P10 (Max) recoverable volumes.

TECHNICAL OPPORTUNITY ASSESMENT AND RECOMMENDED DEVELOPMENT STRATEGY



Development Concept	Technical Target	Est. CAPEX	Est. OPEX	Gross Revenue	Net Cash Flow	Viability Ranking
Group 6: P8 with W7 & W11	NW Lobe Complex	\$160M	\$155M	\$975M	+\$660.0M	1 - Highly Viable (Anchor Project)
Group 4: P3 with W5 & W6	Northern segment optimization	\$0M	\$82M	\$525M	+\$443.0M	2 - Highly Viable (Low Capital Risk)
Group 1: P2+P9 with W3, W4 & W11	Southern sweep expansion	\$160M	\$92M	\$450M	+\$198.0M	3 - Viable (Good Return)
Group 3: P6, P7 with W9	Isolated central Block producer-injector	\$240M	\$78M	\$487.5M	+\$169.5M	4 - Viable (Capital Intensive)
Group 5: P4 with W8 & A1	Marginal flank development	\$0M	\$15M	\$75M	+\$60.0M	5 - Marginal (Sensitive to OPEX)
Group 2: P1, A3 with W1	High-risk transition compartment	\$0M	\$4M	\$37.5M	+\$33.5M	6 - Marginal (Low Yield)

FDP OPPORTUNITY ASSESSMENT & RANKING

Total Portfolio Value: Six development concepts were evaluated, yielding a combined estimated Net Cash Flow (NCF) of \$2.55 Billion.

Tier 1: Anchor Projects (Highly Viable): Groups 6 & 4 are the primary value drivers. Together they generate over \$1.1 Billion in NCF, with Group 6 acting as the flagship High Net Channel infill.

Tier 2: Secondary Targets (Viable): Groups 1 & 3 provide strong additional returns (+\$367.5M NCF) but are highly capital intensive, requiring a combined \$400 M in CAPEX.

Tier 3: Marginal Assets (High Risk): Groups 5 & 2 offer minimal yields (+\$93.5M NCF combined). Due to their high sensitivity to OPEX and low capital efficiency, these transition and flank compartments are recommended for suspension.

RECOMMENDED EXECUTION & RISK STRATEGY

Phase 1: De-Risk (Information First): Deploy \$20M for high-resolution 3D Seismic. Goal: Resolve stratigraphic uncertainty before committing heavy drilling capital, protecting Expected NPV from downside dry-hole risks.

Phase 2: Core Execution (Anchor Projects): Prioritize high-confidence, high-yield infill targets (like Group 6). Goal: Secure baseline cash flow early to fund subsequent field operations.

Phase 3: Secondary Expansion: Execute viable secondary producer-injector pairs. Goal: Maximize sweep efficiency while strictly monitoring the Voidage Replacement Ratio (VRR ~1.0) to control lifting OPEX.

Phase 4: Suspension (Risk Mitigation): Halt development on marginal, low-yield compartments. Goal: Preserve the platform's limited fluid-handling capacity for the most profitable wells and protect the portfolio against downside oil price volatility.

Execution Phase	Strategic Action	Uncertainty & Risk Mitigation Justification
Phase 1: De-Risking	Execute \$20M 3D Seismic & Attribute Mapping campaign.	Resolves geological uncertainty (High Net Channel vs. Lobe Complex boundaries) <i>before</i> committing \$100M+ to drilling CAPEX. Protects ENPV from downside P90 dry-hole risks.
Phase 2: Core Development	Drill/Execute Group 6 and optimize Group 4 .	Capitalizes on areas with highest technical confidence (bypassed oil clearly visible on 4D). Secures \$1.1B+ in baseline net cash flow to fund future operations.
Phase 3: Secondary Infill	Drill/Execute Group 1 and Group 3 .	Expands the sweep efficiency. Risk is managed by actively monitoring the Voidage Replacement Ratio (VRR ~1.0) to prevent early water breakthrough and control lifting OPEX.
Phase 4: Suspension	Suspend or divest from Groups 2 and 5 .	Avoids allocating limited topside fluid-handling capacity to low-yield wells. Prevents these marginal assets from generating negative cash flow if oil prices drop.

Thank
you